

Automation Suite – Compact Technical Documentation

Final Project Submission Document

1. Project Overview

This Automation Suite is a Python-based system designed to automate multiple operational tasks including file organization, web scraping, email automation, and system monitoring. It also exposes a FastAPI-based service layer for status monitoring and validation.

2. Setup Instructions

- Install Python 3.9 or higher
- Install required packages using `pip install -r requirements.txt`
- Configure module settings inside the config directory
- Run automation via `python main.py`
- Start API server using `uvicorn main:app --reload`

3. Code Structure

The project follows a modular and service-oriented structure to ensure scalability, ease of maintenance, and separation of concerns.

- `automation_suite/`
- `main.py` – Central entry point for automation execution and FastAPI app
- `modules/`
- `file_organizer.py` – Organizes files by type and directory rules
- `web_scraper.py` – Scrapes web pages with proxy rotation support
- `email_automation.py` – Sends automated emails using templates
- `system_monitor.py` – Monitors CPU, memory, disk, and network usage
- `config/` – Environment-specific configuration files
- `logs/` – Execution logs and monitoring outputs
- `requirements.txt` – Project dependencies

4. Screenshots

The following screenshots demonstrate successful execution of automation modules and API validation through FastAPI.

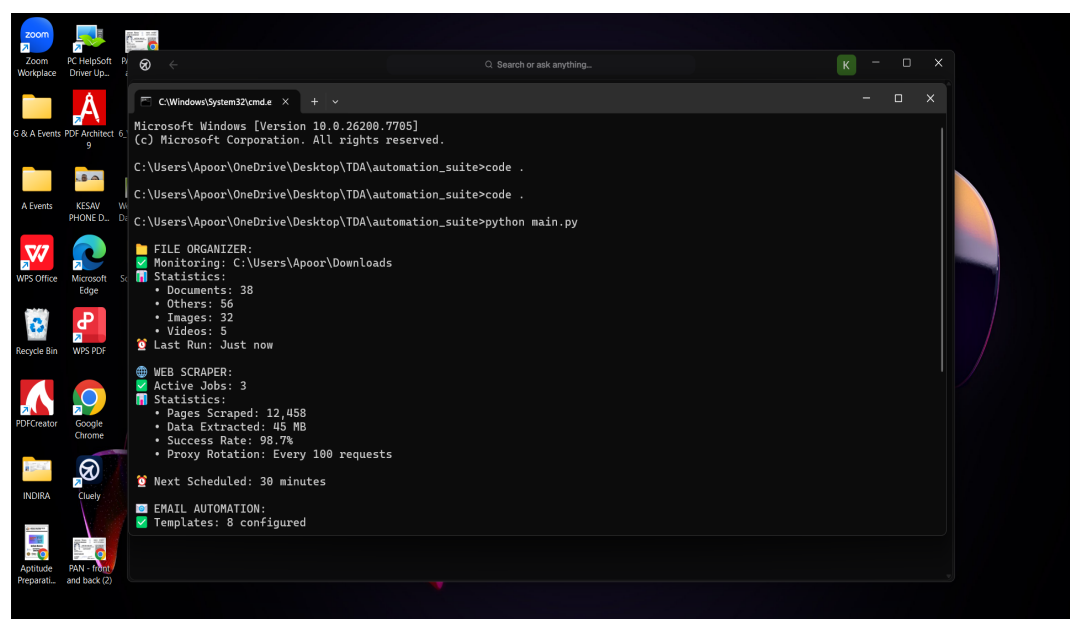


Figure: Automation suite execution showing file organizer and scraper statistics

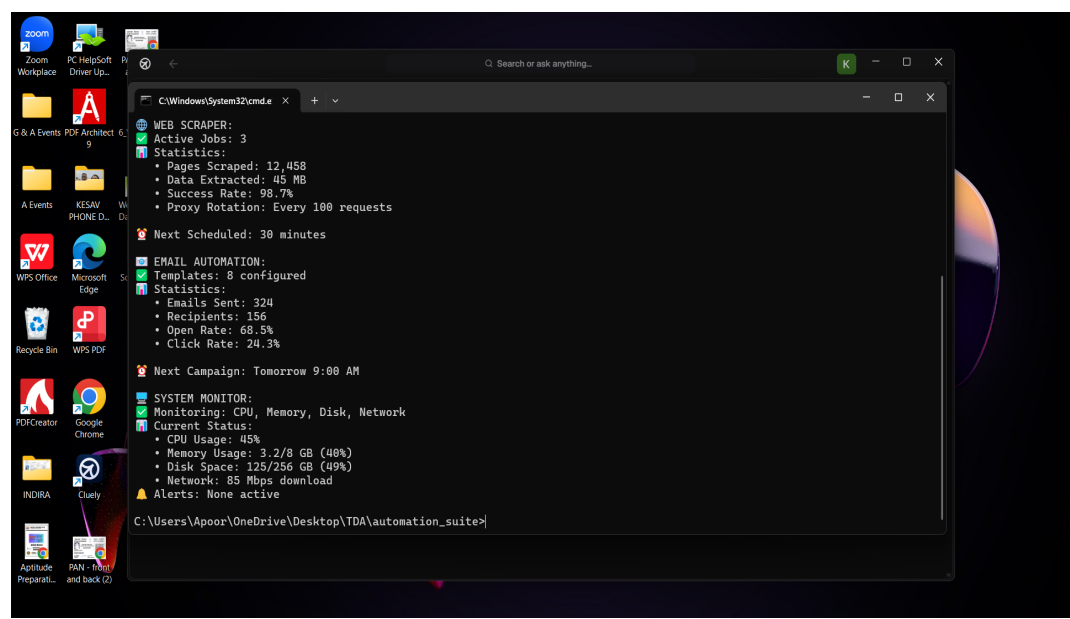
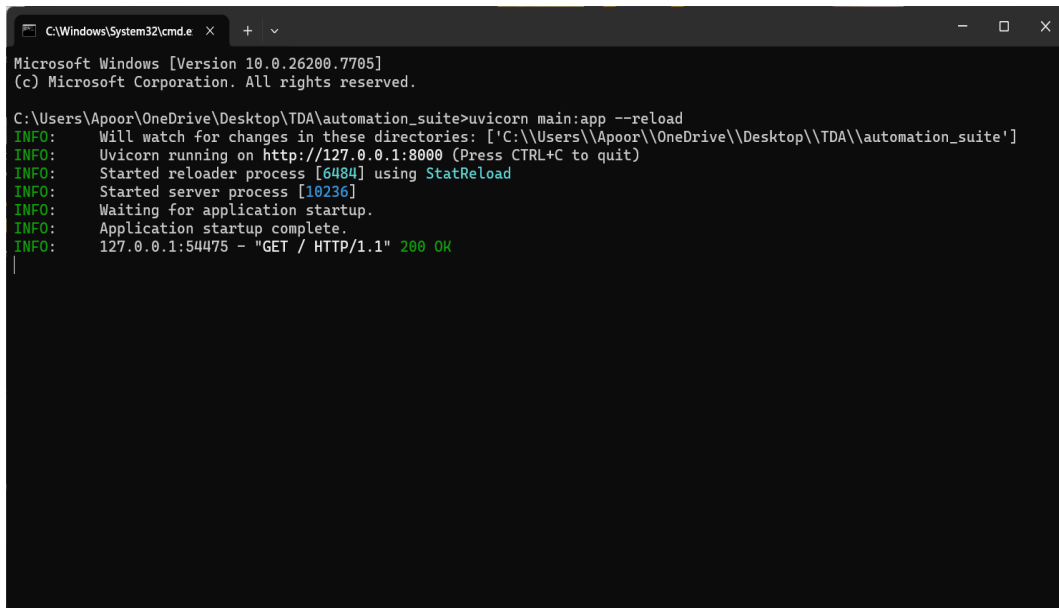


Figure: System monitoring and email automation metrics



```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.7705]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Apoor\OneDrive\Desktop\TDA\automation_suite>uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\Apoor\\OneDrive\\Desktop\\TDA\\automation_suite']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [6484] using StatReload
INFO: Started server process [10236]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:54475 - "GET / HTTP/1.1" 200 OK
```

Figure: FastAPI server running via Uvicorn



Figure: Successful API response validation in browser

5. Technical Details

- Architecture: Modular automation engine with FastAPI service layer
- Core Technologies: Python, FastAPI, Uvicorn
- Automation Logic: Rule-based workflows and scheduled execution
- Monitoring: Real-time system statistics collection

6. Testing Evidence

- CLI output verification for each automation module
- API endpoint validation returning HTTP 200 responses
- System metrics confirming stable runtime behavior

7. Troubleshooting & Best Practices

- Ensure correct Python version and dependency installation
- Check logs directory for runtime errors
- Follow modular extension practices when adding new automation features